PRANAV LAL K B

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Summary

Python developer focused on AI, machine learning, and backend engineering. Experienced in building intelligent applications, REST APIs, and data-driven services using Python, FastAPI, Django, PostgreSQL, and modern AI tooling. Interested in applying software engineering and Generative AI techniques to real-world products.

Technical Skills

Languages: Python, JavaScript
Frameworks: FastAPI, Django, Next.js, TensorFlow
AI/ML: RAG, LangGraph, LangChain, ChromaDB
Databases: PostgreSQL
Backend/Tools: Node.js, AWS, Docker, Git, Postman

Experience

Software Engineer Trainee – Digicyfr *Nov 2025 – Mar 2026*

- Developed and customized Odoo ERP modules, business workflows, and internal features.
- Developed business websites and web application features using Next.js.
- Assisted with debugging, testing, API integration, and cross-system feature implementation.

Fullstack Developer Intern – Odin Tech *Jun 2025 – Sep 2025*

- Contributed to production-oriented web applications, backend logic, and REST API integration.
- Built backend services with FastAPI and PostgreSQL for data storage and API-driven communication.
- Developed frontend components using Next.js and supported UI enhancements and database optimization.

Projects

CodeRAG — AI-Powered Multi-Repository Code Search Platform
LangGraph, FastAPI, ChromaDB

- Built a multi-repository RAG platform indexing GitHub codebases for conversational, context-aware technical Q&A.
- Implemented AST-aware code chunking with dense and BM25 retrieval for code search and question answering.
- Designed repository-specific ChromaDB collections with reranking and LangGraph-based conversational workflows.

Fisherman Safety Advisory Agent
Python, LangGraph, FastAPI, Next.js, Gemini API

- Developed a LangGraph-based agentic AI pipeline integrating six live marine data feeds from Open-Meteo, IMD, and INCOIS to generate personalized GO / CAUTION / NO GO safety advisories.
- Implemented a safety-first architecture with deterministic risk scoring, fishing-zone ranking, LLM failover, Malayalam translation, and neural text-to-speech.

Education

Vidya Academy of Science and Technology – MCA *2023 – 2025, CGPA: 8.3*

Christ College (Autonomous) – BCA *2020 – 2023, CGPA: 7.6*

Certifications & Achievements

- Artificial Intelligence and Machine Learning Developer – ASAP Kerala
- AWS Cloud Practitioner Essentials, Amazon Web Services (AWS)
- District-Level Winner – Young Innovators Programme (YIP)